

2001 MPC Replication Package

Replication code for “Using Macro Counterfactuals to Assess Plausibility: An Illustration using the 2001 Rebate MPCs” by Jacob Orchard, Valerie Ramey, and Johannes Wieland

Tested on MAC and Linux using STATA version 18.5 and Python 3.13.

The project is set up to work in a MAC or LINUX environment (UNIX more generally). We provide instructions for Windows in the Github repository for the paper Micro MPCs and Macro Counterfactuals: The Case of the 2008 Rebates.

Permissions

You can use our code with proper attribution.

Please cite as:

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Statement about Rights

The authors of the manuscript have legitimate access to and permission to use the data used in this manuscript. The authors of the manuscript have documented permission to redistribute/publish the data contained within this replication package.

To Run Entire Project

You will need your own FREDKEY and BEA keys to download the source data. Place the FREDKEY in line 34 of `MPC/forecasting/code/build_forecast_data.do` and place the BEA key in line 14 of `MPC/downloaddata/code/pcefromBEA.py` and line 16 of `MPC/downloaddata/code/pull_pce_detail.py`.

We use `make` to run the entire project. `cd` into the base directory and run the following commands in your terminal:

1. `make install`
2. `make venv`
3. `make`

The first command builds the Python virtual environment, the second command executes the project.

Once `make` executes successfully, the paper figures and tables are available in the folder `_finaltablesandfigures/output`.

Expected Running Time

Around 2 hours.

Required Packages

For a list of required python packages see `requirements.txt` (running ‘`make install`’ should automatically install the required packages). In addition to the packages in `requirements.txt`, some users have had to install `fastparquet` directly.

STATA packages:

```
ssc install did_imputation
```

```
ssc install ranktest
```

```
ssc install ivreghdfe
```

For `ivreghdfe`, it may be useful to install the latest version from github to avoid a “last estimates not found” error (see this help article [here](#)). We are grateful to the data editor for pointing out this issue.

Order of Tasks to Create Final Output

This project is divided into a series of subfolders that execute all of the tasks leading to final output beginning with `downloaddata` and ending with `_finaltablesandfigures`. Each subfolder contains both a code directory and, once-executed, input and output directories. The `makefile` in the code folder documents how the inputs are converted in the outputs for the task. The input directory will have symbolic links to output from previous tasks, while the output directory will include all of the output used by subsequent tasks. The final output for the paper is mostly created in the `forecasting`, `psmjregressions`, `model`, and `narrative` subfolders.

The `makefile` in the main folder shows the order of execution of the subfolders. The graph below shows the relationship between tasks.

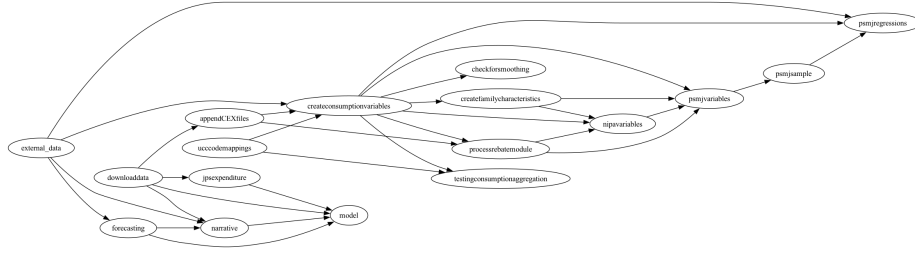


Figure 1: Task Flow

Description of Subfolders:

- `downloaddata`: Downloads CEX data from the BLS and saves as zip files. Also downloads PCE data from the BEA.
- `appendCEXfiles`: Unzips the downloaded CEX data, appends household info (fmli), expenditure (mtbi) and tax rebate (TAX) files together.
- `uccodemappings`: Creates code mappings from CEX ucc codes to larger JPS or PCE consumption aggregates
- `createconsumptionvariables`: Uses the CEX data and the uccodemappings to create consumption variables for each household
- `testingconsumptionaggregation`: Tests that aggregated consumption from the MTBI files is equal to household consumption in the FMLI files
- `processrebate module`: Creates rebate amount variable and indicator for each household
- `createfamilycharacteristics`: Creates other household level variables
- `nipavvariables`: Merges relevant PCE household level variables with household characteristics
- `psmjvariables`: Merges consumption based on JPS definition with household characteristics
- `psmj sample`: Creates final sample of households following JPS paper
- `psmj regressions`: Empirical regressions and tables (Tables 4, 5, 6, and appendix Table B.1)
- `jpsexpenditure`: Uses the JPS definition of consumption to create a time series of aggregate nondurable expenditure.
- `survey_prof_fore`: Downloads and cleans data from the Survey of Professional Forecasters (Figure 5)
- `forecasting`: Creates our own forecasts from section 5 of the paper (Figure 5)
- `narrative`: Creates narrative figures (Figures 1, 2 and 3)
- `model`: Solves model and produces Tables 1, 2, 3, Figure 4, and appendix Figure C.1.
- `symlink_graph`: Creates chart showing input-output dependencies above
- `_finaltablesandfigures`: Tables and figures used in the paper will be copied here.

Creating individual outputs

The following table lists which figure is generated by which final task. One can generate one figure at a time by going into the corresponding **task/code** directory and running **make**.

Figure/Table	Figure/Table Filename	Task Directory
Figure 1	fig_cy_pres.eps	narrative
Figure 2	fig_rebate_ndisp01.eps	narrative
Figure 3a	fig_cy_pres01.eps	narrative
Figure 3b	fig_jpscy_pres01.eps	narrative
Figure 4a	Real_JPSNondurablesfc_micro_nondurablesonly2001.eps	narrative
Figure 4b	Real_JPSNondurablesfc_GE_nondurablesonly2001.eps	narrative
Table 1	calibrationnondurablesonly2001gamma.tex	model
Table 2	mpcsgammanondurablesonly2001.tex	model
Table 3	Real_JPSNondurables_911_GE_nondurablesonly2001.tex	model
Figure 5a	SPF_dist_0102_wactual.pdf	survey_prof_fore
Figure 5b	fig_forecasts2001_jpsnondur.eps	forecasting
Table 4A	Table_2001_NDcompare.tex	psmjregressions
Table 4B	Table_2001_NDcomparerbtonly.tex	psmjregressions
Table 5	bea_all.tex	psmjregressions
Table 6A	Table_2001_NDcompare_wlag.tex	psmjregressions
Table 6B	Table_2001_NDcompare_wlagrbtonly.tex	psmjregressions
Appendix	coefsall_prettytable.xlsx	psmjregressions
Table B.1		
Appendix	Real_JPSNondurablesfc_micro_nondurablesonly2001_appendix.eps	narrative
Figure C.1		

For data that may be revised in the future, such as aggregate data downloaded from FRED, we include a plain text version of the data in the subfolder **report**.

Data Citation

This project code downloads data from the Bureau of Labor Statistics (BLS), the Federal Reserve Bank of Philadelphia (the SPF surveys), and the Bureau of Economic Analysis (BEA). The user of these replication files will be under the license requirments of these files when they run the replication code.

Downloaded Data Citation

Bureau of Economic Analysis (2000-2002). ‘National income and product accounts’, US Department of Commerce

Bureau of Labor Statistics (2000-2002). ‘Consumer expenditure survey’, US Department of Labor.

Federal Reserve Bank of Philadelphia (2001). ‘Second quarter 2001 survey of professional forecasters’, Survey of Professional Forecasters.

External datasets

All datasets that are not downloaded directly by the code are included in the folder `external_data`. There are also .csv versions of all of the below data files.

Name	Source	Citation	License
rebates.xlsx	Authors and Slemrod Shapiro, Sahm, and Slemrod Table 6 and Shapiro and Slemrod	Shapiro, and Slemrod (2003), Sahm, Shapiro, and Slemrod (2012)	Creative Commons and GNU General Public License v3.0*
JPS_consumption	Authors and BEA	Orchard, Ramey, Wieland (Forthcoming) and BEA (2000-2002)	Public domain and GNU General Public License v3.0
ce-pumd-interview-diary-dictionary.xlsx	BLS	BLS (2024)	Public domain
BEA_labels.xls	Authors	Orchard, Ramey, Wieland (Forthcoming)	GNU General Public License v3.0

*Authors use “GNU General Public License v3.0” while Sahm, Shapiro, and Slemrod (2012) use the Creative Commons license. The creative commons license is copied in this replication folder under `license_sss`.

Bureau of Economic Analysis (2000-2002). ‘National income and product ac-

counts', US Department of Commerce

Bureau of Labor Statistics (2000-2002). 'Consumer expenditure survey', US Department of Labor.

Orchard, Jacob, Valerie A. Ramey, and Johannes F. Wieland. Using Macro Counterfactuals to Assess Plausibility: An Illustration using the 2001 Rebate MPCs. *Economic Journal*, Forthcoming.

Sahm, Claudia R, Matthew D Shapiro, and Joel Slemrod, 2012. "Check in the mail or more in the paycheck: does the effectiveness of fiscal stimulus depend on how it is delivered?" *American Economic Journal: Economic Policy* 4(3): 216–50.

Shapiro, Matthew D and Joel Slemrod, 2003b. "Did the 2001 tax rebate stimulate spending? Evidence from taxpayer surveys." *Tax policy and the economy* 17: 83–109.